Math 534H Lec 15

Stability ~~Stability~~ of solutions to Heat/Diffusion equation

Consider two solution to the IVBP for H/D equation

$$\begin{cases} \partial_t u_1 - k\partial_{xx} u_1 = 0, & \boxed{x\in[0,L] \quad t>0} \\ u_1(x,0)=\phi_1(x) \\ u_1(0,t)=g_1(t), & \boxed{u_1(L,t)}=h_1(t) \end{cases}$$

$$\begin{cases} \partial_t u_2 - k\partial_{xx} u_2 = 0 \\ \boxed{u_2(x,0)=\phi_2(x) \leftarrow \text{初值}} \\ u_2(0,t)=g_2(t) \qquad u_2(L,t)=h_2(t) \leftarrow \text{边界值} \end{cases}$$

Assume: $\phi_1 \xrightarrow{\text{closeness}} \phi_2 \qquad g_1 \xrightarrow{\text{closeness}} g_2$

$u_1 \xrightarrow{\text{close}} u_2$

Consider $\quad w = u_1 - u_2$

$$\begin{cases} \partial_t w - k\partial_{xx} w = 0 \\ w(x,0) = \phi_1(x) - \phi_2(x) \\ w(0,t) = g_1(t) - g_2(t) \qquad w(L,t) = h_1(t) - h_2(t) \end{cases}$$

By max/min principle imply:

$$\min_{(x,t)\in I}\{w(x,t)\} \le \underline{w(x,t)} \le \max_{(x,t)\in I}\{w(x,t)\}$$

$(x,t)\in R$

Infact ① $-\max_{(x,t)\in I}\{|w(x,t)|\} \le \max_{(x,t)\in R}\{w(x,t)\} \le \max_{(x,t)\in I}\{|w(x,t)|\}$

$\Downarrow$

② $\max_{(x,t)\in R}\{|u_1(x,t)-u_2(x,t)|\} \le \max_{(x,t)\in I}\{|w(x,t)|\}$

$= \max_{(x,t)\in I}\{|\phi_1(x)-\phi_2(x)|, |g_1(t)-g_2(t)|, |h_1(t)-h_2(t)|\}$

热/扩散方程初值问题 $\longrightarrow$
解的稳定性

如果初值和边界值只改一点点，那么对应的解也只会改一点点

两个解 u_1, u_2 对应于纠结
接近的数据.

$\phi_1 \simeq \phi_2,\ g_1 \simeq g_2,\ h_1 \simeq h_2$

整个区域中，w 不会比边界的最小值小，也不会比边界上的最大值大

$$\max_R |w(x,t)| \le \max_I |w(x,t)|$$

$$\max_R |u_1(x,t)-u_2(x,t)| \le \max_I\{|\phi_1-\phi_2|, |g_1-g_2|, |h_1-h_2|\}$$

在时空区域里，两解之间的差异，不会超过它们的初值和边界值差异的最大值（最大值）

一致的/最大值意义下

从上页的 uniform stability (一致稳定性),
过渡到 energy stability (能量稳定性)

$\boxed{\text{Uniformly}}$

特别数据在最大范数下接近，则解也在最大值范数下接近

Hence if the maximum of $|\phi_1(x)-\phi_2(x)|$ and the maximum of $|g_1(t)-g_2(t)|$ and of $|h_1(t)-h_2(t)|$ are small then maximum of $|u_1(x,t)-u_2(x,t)|$ is also small.

$E[u](t) = \frac{1}{2}\int_0^L |u(x,t)|^2 dx$

measure stability ~~by~~ / square stability by the energy functional.

热方程的能量是随着时间递减至为不增大

Claim $E[u](t)$ is decreasing.

$$0 \le E[u](t) \le E[u](0) \qquad \forall u_{(\text{solution})} \quad t>0$$

解在时刻t的能量，不超过初始时刻的能量

The energy of the solution is controlled by $E[|\phi_1-\phi_2|]$

$$E[|\phi_1-\phi_2|] = E[w](0)$$

In fact, we will proof the energy for the H/D equation is making bate:

$$E[u](t) = \frac{1}{2}\int_0^L |u(x,t)|^2 dx$$

and $\frac{dE}{dt} \le 0$ so energy decrease. $w(x,0)=\phi_1(x)$

$E[w](t)$ $\qquad$ $E[w](0)$, $\phi_1-\phi_2(x)$

$\boxed{\phi(x)\equiv 1, t=0}$ 各方面能量

$$\int_0^L |u_1(x,t)-u_2(x,t)|^2 dx \le \underbrace{\int_0^L |\phi_1(x)-\phi_2(x)|^2 dx}_{\text{number}}$$

This type of stability is called in the square integral sense.

称为平方积意义下的稳定性，也称是L_2稳定性

Resistence Term / Forcing term

我们证明热方程能量递减

Energy functional for H/D equations.

$$0 = u_t - k u_{xx}$$

$$u\cdot 0 = u_t u - k u_{xx} u$$

$$\boxed{0 = u u_t - k u u_{xx}}$$

下文的w等价于u

$$\begin{cases} w_t - k w_{xx} = 0 \\ w(x,0) = 0 \\ w(0,t) = 0 \\ w(L,t) = 0 \end{cases} \qquad x \in [0,L]$$

$$\begin{cases} 0 = w_t - k w_{xx} \\ w\cdot 0 = (w_t - k w_{xx}) w = 0 \\ 0 = w w_t - k w w_{xx} \end{cases}$$

$$0 = \int_0^L w w_t \, dx - k\int_0^L w w_{xx} \, dx$$

$$\int_0^L \frac{1}{2}\frac{d}{dt}(w^2)\, dx$$

$$-k\underbrace{\left[w w_x\right]_{x=0}^{x=L}}_{=0} - k\int_0^L (w_x)^2 dx$$

Hence

$$\frac{1}{2}\frac{d}{dt}\int_0^L |w(x,t)|^2 dx = \underbrace{-k\int_0^L (w_x)^2 dx}_{\le 0}$$

$$E[w](t) = \frac{1}{2}\int_0^L |w(x,t)|^2$$

$$\frac{dE}{dt} \le 0 \qquad \boxed{\text{decrease.}}$$

用上一页的 energy decrease 来证明唯一性

Uniqueness: u_1 and u_2 ← solution (IVBP)

$\Downarrow$

$\underline{w = u_1 - u_2}$ solve. IVBP

$\downarrow$ Energy decrease.

$E[w](t) \le E[w](0)$

$\le \frac{1}{2}\underbrace{\int |w(x,0)|^2 dx}_{=0} \Leftarrow w(x,0) = 0$

能量转一定非负

$\downarrow$

$0 \le E[w](t) \le 0$

$\Downarrow$

$0 \le \underbrace{\int_0^L |w(x,t)|^2 dx}_{0} \le 0$

$\Downarrow$

~~$w(x$~~ $\Downarrow$

$\underline{w(x,t) = 0}$ $\forall x.\ \forall t$

$\Downarrow$

$\boxed{w = 0 = \overset{u_1 - u_2}{\cancel{u_1 - u_2}}} \Rightarrow \boxed{u_1 = u_2}$ ← Uniqueness

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$$v_t(x_0, T) = \lim_{\delta \to 0^+} \frac{v(x_0, T) - v(x_0, T-\delta)}{\delta} \ge 0$$

$v_t \ge (x_0, T) \ge 0$

$\Downarrow$

$v_t - k v_{xx} \ge 0$, 矛盾